File permissions in Linux

Project description

The company wants to update the access and reading permissions for certain files on its system because they do not adhere to the principle of least privilege.

Check file and directory details

To see details on files in a directory we can use the command “ls -la”

```
researcher2@6954146e11b2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  7 14:01 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun  7 14:37 ..
-rw--w---- 1 researcher2 research_team   46 Jun  7 14:01 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun  7 14:01 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jun  7 14:01 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jun  7 14:01 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun  7 14:01 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun  7 14:01 project_t.txt
```

Describe the permissions string

1. **First Character:**
   - d: Indicates a directory.
   - -: Indicates a regular file.
2. **Next Three Characters (User/Owner Permissions):**
   - r: Read permission for the user who owns the file.
   - w: Write permission for the user who owns the file.
   - x: Execute permission for the user who owns the file.
3. **Next Three Characters (Group Permissions):**
   - r: Read permission for users who are members of the file's group.
   - w: Write permission for users who are members of the file's group.
   - x: Execute permission for users who are members of the file's group.
4. **Last Three Characters (Others Permissions):**
   - r: Read permission for users who are not the owner and not in the group.
   - w: Write permission for users who are not the owner and not in the group.
   - x: Execute permission for users who are not the owner and not in the group.

Change file permissions

To change permissions on a file or directory, we can use the command 'chmod'. The enterprise does not allow write access for unauthorized people, so to remove write permissions for others on the file 'project_k.txt', we can use this command: 'chmod o-w project_k.txt'.

```
researcher2@5d738f0f927b:~/projects$ chmod o-w project_k.txt
researcher2@5d738f0f927b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec  2 15:27 ..
-rw--w---- 1 researcher2 research_team   46 Dec  2 15:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec  2 15:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec  2 15:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec  2 15:27 project_t.txt
researcher2@5d738f0f927b:~/projects$
```

This command will remove write permissions for others.

Change file permissions on a hidden file

The hidden file '.project_x.txt' should be readable for the owner and group but not writable. To remove unwanted permissions and add read permission for the group, we can use 'chmod u-w,g-w,g+r .project_x.txt'.

```
researcher2@3213bbc1d047:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@3213bbc1d047:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 .
drwxr-xr-x 3 researcher2 research_team 4096 Dec 20 15:36 ..
-r--r----- 1 researcher2 research_team   46 Dec 20 15:36 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Dec 20 15:36 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Dec 20 15:36 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Dec 20 15:36 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec 20 15:36 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Dec 20 15:36 project_t.txt
researcher2@3213bbc1d047:~/projects$
```

Change directory permissions

The directory 'drafts' should be accessible only by the user, so we have to remove the execute permissions for the group. To do this, I run the command 'chmod g-x drafts/'.

```
researcher2@6954146e11b2:~/projects$ chmod g-x drafts/
researcher2@6954146e11b2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun  7 14:01 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun  7 14:37 ..
-rw------- 1 researcher2 research_team   46 Jun  7 14:01 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Jun  7 14:01 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jun  7 14:01 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jun  7 14:01 project_m.txt
-rw-rw---- 1 researcher2 research_team   46 Jun  7 14:01 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun  7 14:01 project_t.txt
```

Summary

I updated the permissions of certain files that did not adhere to the principle of least privilege. The command 'ls -la' allows me to display the current permissions, and 'chmod' applies the necessary changes to ensure that access aligns with the desired configuration.